Project: Image Super-Resolution using Machine Learning

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Course Code: CSE 445

Section: 06

Group No: 04

Week 1 (Feb 5 – Feb 11): Project Setup and Initial Image Selection

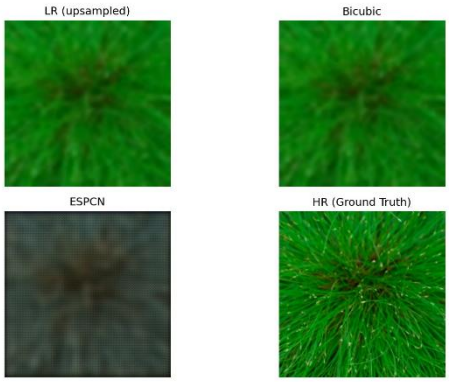
In the first week, my main goal was to lay the groundwork for our project. I was in charge of picking the first 100 images that we would use to train our super-resolution model. I also helped set up the first GitHub repository so that our team would have a central version control system to keep track of all the code and data we added from the start.

Week 2 (Feb 12 – Feb 19): Data Preprocessing and Standardization Pipeline

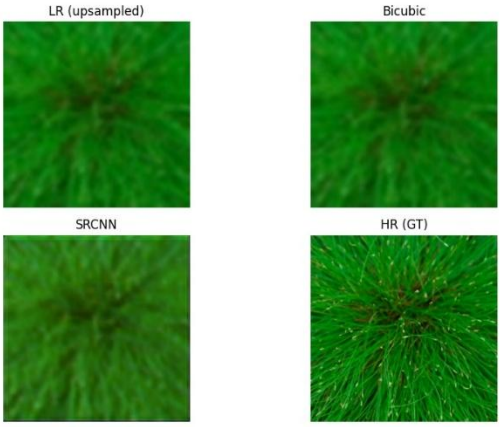
I took full control of the data preprocessing pipeline this week. It was very important for getting our inputs ready. I wrote the **Standardize_images.ipynb** script and added 10 more images to our dataset to make it bigger. My main technical job was to write the code that made all of our high-resolution target images the same size: 256 × 256 pixels. After that, I wrote a script that systematically turned these 256 × 256 images into 64 × 64 low-resolution inputs. During this time, I also took care of the repository operations and was able to successfully review and merge "Pull Request #1" and "Pull Request #2" into the main branch.

Week 3 (Feb 20 – Mar 1): Model Exploration and Pipeline Testing

I proceeded to the machine learning architecture after the datasets were prepared. For the entire week, I looked into super-resolution models that would work with our particular 64 × 64 to 256 × 256 pipeline. Because of its efficiency, I first attempted to implement the **ESPCN** architecture, but I encountered several significant obstacles. Resolving spatial dimension mismatches was the largest challenge. In order to reshape the tensor back into an RGB image, **ESPCN** uses a **PixelShuffle** layer that strictly requires exactly 3 × 42 = 48 channels. Tensor shape errors during the forward pass resulted from even the smallest misconfiguration in the previous convolutional layers. Furthermore, considering our small dataset, the training process was rather erratic. After testing and



discussing these problems with the team, I realized we needed to pivot our approach to establish a more reliable baseline for our structural evaluations.

Week 4 (Mar 2 – Mar 8): Architecture Restructuring to SRCNN

By formally implementing the switch from the **ESPCN model** to the traditional **SRCNN** (Super-Resolution Convolutional Neural Network) architecture, I reorganized the code of our model to enhance our upscaling results. Because SRCNN directly maps a **bicubic-upsampled** low-resolution image to the high-resolution target, I made the switch. This improves our Peak Signal-to-Noise Ratio (PSNR) evaluation metric and makes the optimization process extremely stable. I successfully established the main model we will use for our final evaluations by setting up the network to process the upsampled inputs and committing the final code ("Change ESPCN Model to SRCNN Model") straight to the main repository.